

MTH6134 Practical Week 3

1 Exponential distribution

Let $Y_i, 1 \leq i \leq n$, be independent random variables with probability density given in generalized linear model form by

$$f(y_i; \theta_i, \phi) = \exp \left(\frac{y_i \theta_i - b(\theta_i)}{a(\phi)} + c(y_i, \phi) \right),$$

with $\theta_i = \theta_i(\eta_i)$ a function of the linear predictor

$$\eta_i = x_i^T \beta = (x_{i,1}, \dots, x_{i,p})(\beta_1, \beta_2, \dots, \beta_p)^T.$$

Suppose Y_i has a $\text{Exp}(\lambda_i)$ distribution with mean $E(Y_i) = 1/\lambda_i$. Find the natural parameter and $a(\phi)$ and calculate b as a function of the natural parameter. Calculate the canonical link function. What practical difficulty arises when using the canonical link in this instance?

Answer:

Here

$$f(y_i; \theta_i, \phi) = \lambda_i \exp(-y_i \lambda_i).$$

This is

$$f = \exp(-y_i \lambda_i + \log(\lambda_i)) = \exp(y_i \theta_i + \log(-\theta_i))$$

if the natural parameter is $\theta_i = -\lambda_i$. In this setting $a(\phi) = 1$, $c = 0$ and $b(\theta_i) = -\log(-\theta_i)$.

The mean is $\mu_i = 1/\lambda_i$ so $\mu_i = -1/\theta_i$.

If we want $\theta_i = \eta_i$ (for the canonical link function) then we should take $\mu_i = -1/\eta_i$, so in this case the CLF is $g(\mu_i) = -1/\mu_i$.

Since $\mu_i = -1/\eta_i$, with $-\infty < \eta_i < \infty$, this doesn't guarantee $\mu_i > 0$, so we may need to constrain $\eta_i > 0$ with consequences for β .

Note that in R and elsewhere the canonical link function for $\text{Exp}()$ (and $\text{Gamma}()$) is (re)defined to be $g(\mu_i) = 1/\mu_i$ and the minus-sign is carried through straightforwardly.

2 Normal distribution

Show that the normal distribution is a member of the exponential family of distributions, use the properties of the exponential family pdf to obtain the mean and variance of a normal random variable and the variance of the linear regression coefficients when the canonical link is used.

Answer :

$$\begin{aligned} f(y; \mu, \sigma^2) &= \frac{1}{\sqrt{2\pi\sigma^2}} \exp \left\{ -\frac{(y - \mu)^2}{2\sigma^2} \right\} = \\ &= \exp \left\{ \log \left(\frac{1}{\sqrt{2\pi\sigma^2}} \right) - \frac{(y - \mu)^2}{2\sigma^2} \right\} = \\ &= \exp \left\{ -\frac{1}{2} \log(2\pi\sigma^2) - \frac{1}{2\sigma^2} (y^2 + \mu^2 - 2y\mu) \right\} = \\ &= \exp \left\{ \frac{y\mu - \mu^2/2}{\sigma^2} - \frac{y^2}{2\sigma^2} - \frac{1}{2} \log(2\pi\sigma^2) \right\} \end{aligned}$$

So that $y\theta = y\mu$, $\theta = \mu$, $\kappa(\theta) = \mu^2/2 = \theta^2/2$, $c(y, \phi) = -\frac{y^2}{2\sigma^2} - \frac{1}{2} \log(2\pi\sigma^2)$ and $\phi = \sigma^2$.

The canonical link is $g(\mu_i) = \mu_i$ with $g'(\mu_i) = 1$.

Now $E(Y) = \kappa'(\theta) = \frac{1}{2}2\mu = \mu$ and $\text{Var}(Y) = \phi\kappa''(\theta) = \sigma^2$ while $\text{Vf}(\mu) = \kappa''(\theta) = 1$ so that $W_{ii} = \frac{1}{\sigma^2}$ which gives $I = X^T X \frac{1}{\sigma^2}$ and $I^{-1} = \sigma^2 (X^T X)^{-1}$ which is the known expression for the variances of regression coefficients in the linear model.

3 Poisson regression with one covariate

Take a toy dataset $y = (1, 2, 3)$ with covariate $x = (0, 1, 2)$ and canonical log link and linear predictor for the i th observation $\eta_i = \beta_0 + \beta_1 x_i = x_i^T \beta$, and $\mu_i = \exp(\eta_i)$.

1. Write $\ell(\beta)$, show the score is $U(\beta) = X^T(y - \mu)$ with $X = [\mathbf{1}, x]$ and that the expected information is $\mathcal{I}(\beta) = X^T W X$ with $W = \text{diag}(\mu)$.

Answer:

$\ell(\beta) = \sum_i \{y_i(x_i^T \beta) - e^{x_i^T \beta} - \log(y_i!)\}$. Differentiating gives $U(\beta) = \sum_i x_i(y_i - \mu_i) = X^T(y - \mu)$. For Poisson with canonical link, $\text{Var}(Y_i) = \mu_i$, so $\mathcal{I}(\beta) = X^T \text{diag}(\mu) X = X^T W X$.

2. Start IWLS with $\mu^{(0)} = y$ so $\eta^{(0)} = \log y$, $z^{(0)} = \eta^{(0)}$ and $W^{(0)} = \text{diag}(\mu^{(0)})$.
 - (a) Using R, compute $\beta^{(1)} = (X^T W^{(0)} X)^{-1} X^T W^{(0)} z^{(0)}$.
 - (b) Update $\eta^{(1)} = X \beta^{(1)}$, $\mu^{(1)} = \exp(\eta^{(1)})$, $z^{(1)} = \eta^{(1)} + (y - \mu^{(1)})/\mu^{(1)}$, and $W^{(1)} = \text{diag}(\mu^{(1)})$.
 - (c) Compute $\beta^{(2)}$ using $W^{(1)}, z^{(1)}$.

Answer

```
y <- c(1, 2, 3)
x <- c(0, 1, 2)
X <- cbind(1, x)
# Initial values (as in notes)
mu <- y
eta <- log(mu)
z <- eta
W <- diag(mu)
# One iteration
beta1 <- solve(t(X) %*% W %*% X) %*% (t(X) %*% W %*% z)
eta1 <- X %*% beta1
mu1 <- exp(eta1)
z1 <- eta1 + (y - mu1) / mu1
W1 <- diag(as.numeric(mu1))
# Second iteration
beta2 <- solve(t(X) %*% W1 %*% X) %*% (t(X) %*% W1 %*% z1)
# Fit with glm for comparison
fit <- glm(y ~ x, family = poisson)
coef(fit); summary(fit)
# Compare beta2 to coef(fit)
```

Your $\beta^{(2)}$ should be very close to the MLE from `glm()`. The reported standard errors equal the square roots of the diagonal of $(X^T W X)^{-1}$ at the MLE.

4 Binomial distribution

Show that the Binomial distribution with m trials is a member of the exponential family of distributions, use the properties of the exponential family pdf to obtain the mean and variance of a binomial random variable and identify the canonical link.

Answer :

Binomial (with m_i trials) pmf in canonical form:

$$f(y; m, p) = \binom{m}{y} p^y (1-p)^{m-y} = \exp \left(y \log \frac{p}{1-p} - m \log(1+e^\theta) + \log \binom{m}{y} \right),$$

with canonical parameter $\theta = \log(p/(1-p))$ and $\kappa(\theta) = m \log(1+e^\theta)$. Here $\phi = 1$. Then

$$\mu = \kappa'(\theta) = m \frac{e^\theta}{1+e^\theta} = mp, \quad V(\mu) = \kappa''(\theta) = m \frac{e^\theta}{(1+e^\theta)^2} = mp(1-p).$$